

Project#7

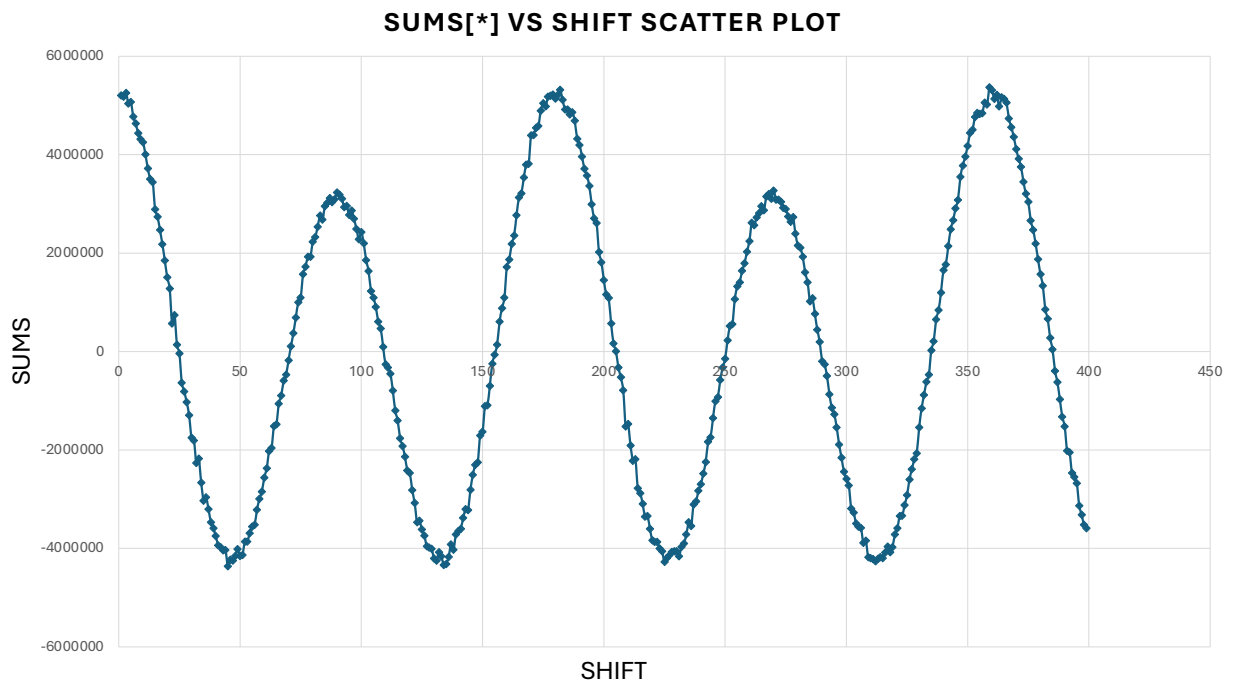
Autocorrelation using MPI

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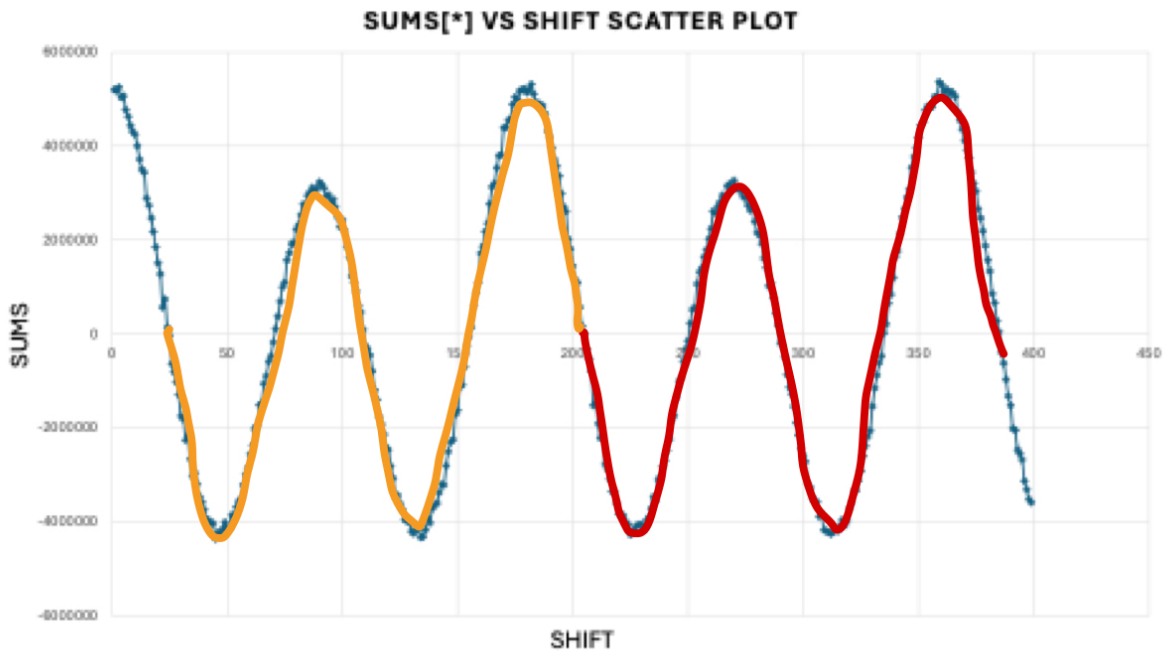
June 11, 2024

1. Scatter Plot



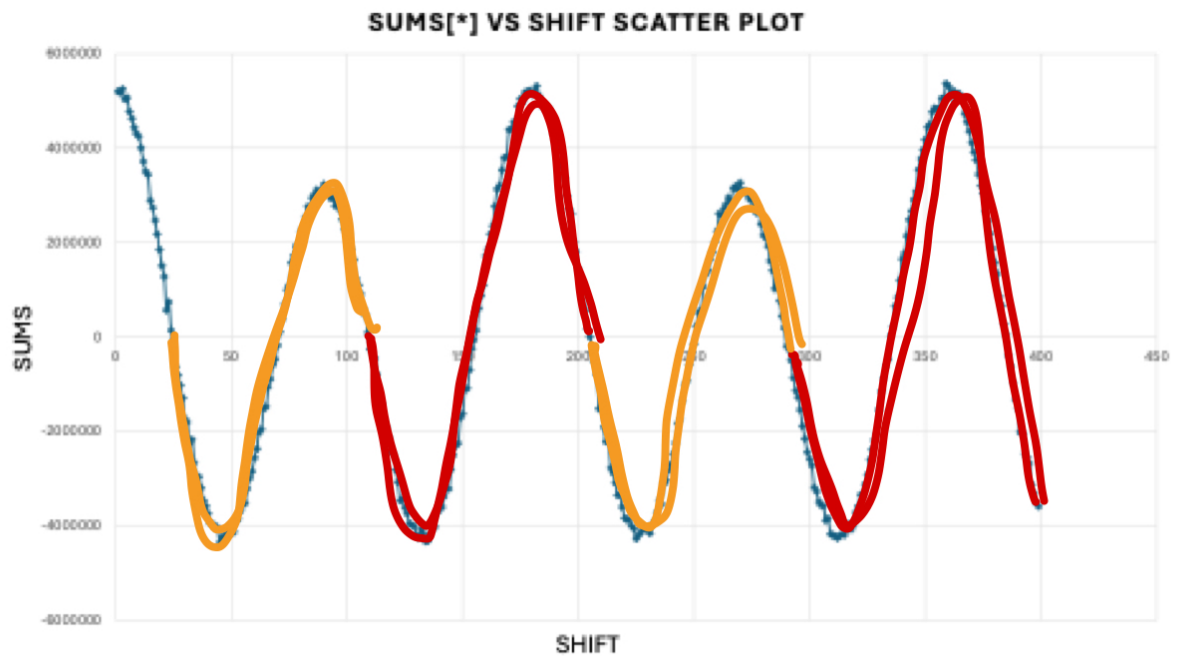
2. Periods of the two-secret sinewave

- To identical patterns that are repeated was observed in the graph.
- Highlighted with Orange and Red color, making it clear.



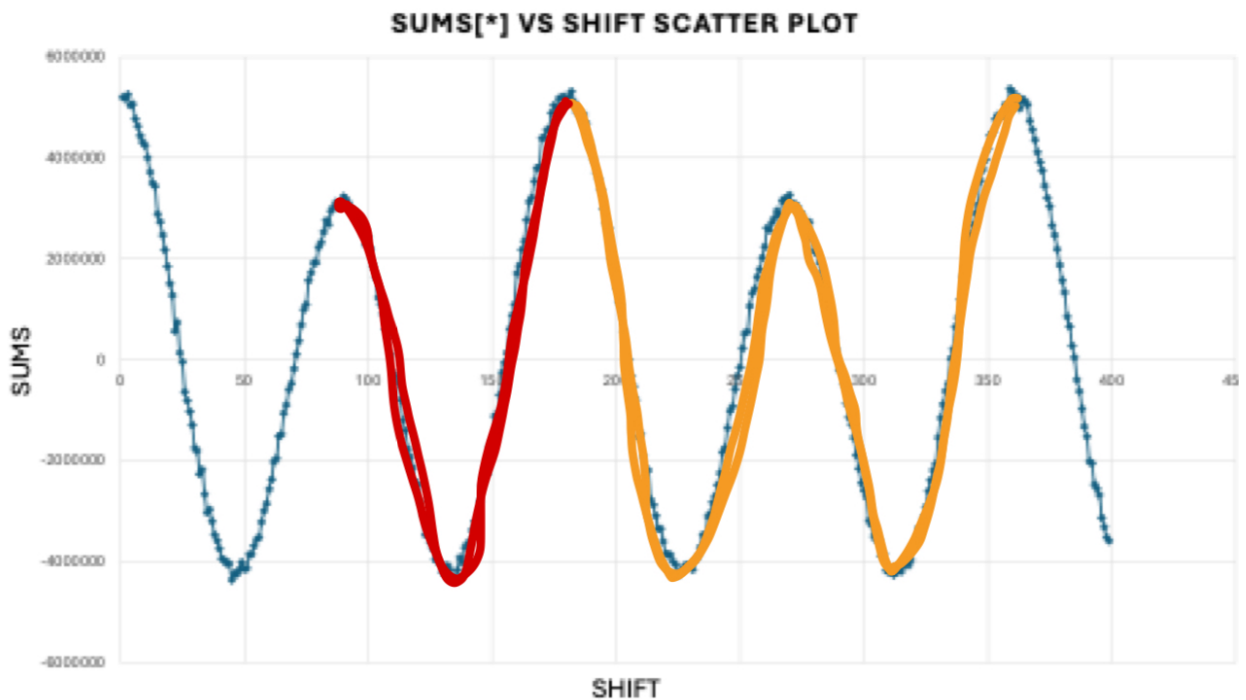
- To find the period, I need to confirm the starting and the ending points of each graphs
- 1st Graph
 - Starting point
 - 24, 139539.62
 - 25, -37356.56
 - Since this part is where the scattered plot jumps to positive integer to negative integer and 24's value is closer to 0, I will set starting point as 24
 - Ending point
 - 205, 7453.70
 - 206, -324292.00
 - With the same reason, I will set ending point 205
- 2nd Graph
 - Starting point
 - 205, 7453.70
 - 206, -324292.00
 - Since the right next to ending point of the 1st graph is the starting point. I will set 206 as a starting point.
 - Ending point
 - 385, 42614.12
 - 386, -390838.56
 - With the same reason, I will set ending point 385
- 1st Graph period = $205 - 24 = 181$

- 2nd Graph period = $385 - 206 = 179$
- **It can be said that the period of this sign graph ≈ 180**
- Or maybe the two sine-wave can be interpreted like the picture below



- In this case:
- The small Graph(Orange)
 - Starting point
 - 24, 139539.62
 - 25, -37356.56
 - Since this part is where the scattered plot jumps to positive integer to negative integer and 24's value is closer to 0, I will set starting point as 24
 - Ending point
 - 109, 95176.95
 - 110, -262504.31
 - With the same reason, I will set ending point 109
- The larger Graph(Red)
 - Starting point
 - 109, 95176.95

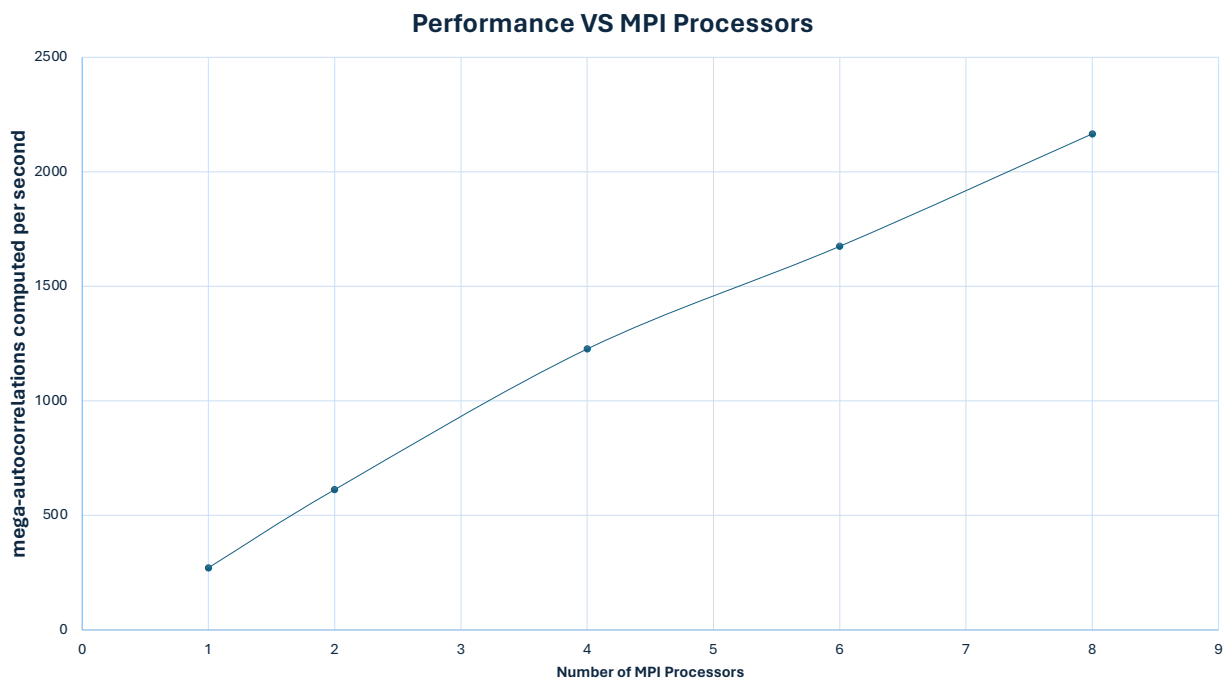
- 110, -262504.31
 - Since the right next to ending point of the 1st graph is the starting point. I will set 109 as a starting point.
- Ending point
 - 205, 7453.70
 - 206, -324292.00
 - I will set ending point 206
- **Small graph period = $109 - 24 = 85$**
- **Larger graph period = $206 - 109 = 97$**
- The period from professor's hint



- In this case:

- Distance between a short peak and the next tall (Red)
 - Starting point
 - 90 , 3233125.50
 - Ending point
 - 182 , 5316669.00
- Distance between successive tall peaks (Orange)
 - Starting point
 - 182 , 5316669.00
 - Ending point
 - 359 , 5366828.00
- **Small graph period = $182 - 90 = 92$**
- **Larger graph period = $359 - 182 = 177$**

3. Performance vs Number of Processors Graph



4. The patterns in Performance Graph

- The performance increases with the number of MPI processors. However, there's a slight reduction in the rate of increase when moving from 4 to 6 processors. Nonetheless, the performance increase returns to the previous trend when moving from 6 to 8 processors.

5. Analysis of the Performance Graph

- As the number of processors increases, the workload is evenly distributed among them, resulting in enhanced efficiency for each processor. The graph confirms that increasing the number of processors boosts performance.
- The slight reduction of the performance graph seems to be due to the busy server at that time.